

Exercise 0

Transaction Summary			
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Install 42 Packages			
Total download size: 102 M			
Installed size: 316 M			
Downloading Packages:			
(1/42): annobin-plugin-gcc-12.69-1.amzn2023.0.1.x86_64.rpm	31 MB/s	974 kB	00:00
(2/42): annobin-docs-12.69-1.amzn2023.0.1.noarch.rpm	2.6 MB/s	93 kB	00:00
(3/42): Lmod-8.4.23-1.amzn2023.0.3.x86_64.rpm	5.2 MB/s	218 kB	00:00
(4/42): gc-8.0.4-5.amzn2023.0.2.x86_64.rpm	2.9 MB/s	105 kB	00:00
(5/42): cpp-11.5.0-5.amzn2023.0.5.x86_64.rpm	34 MB/s	11 MB	00:00
(6/42): gcc-c++-11.5.0-5.amzn2023.0.5.x86_64.rpm	33 MB/s	15 MB	00:00
(7/42): gcc-plugin-annobin-11.5.0-5.amzn2023.0.5.x86_64.rpm	1.5 MB/s	38 kB	00:00
(8/42): gcc-gfortran-11.5.0-5.amzn2023.0.5.x86_64.rpm	40 MB/s	15 MB	00:00
(9/42): glibc-devel-2.34-231.amzn2023.0.5.x86_64.rpm	187 kB/s	33 kB	00:00
(10/42): gcc-11.5.0-5.amzn2023.0.5.x86_64.rpm	37 MB/s	36 MB	00:00
(11/42): glibc-headers-x86-2.34-231.amzn2023.0.5.noarch.rpm	1.3 MB/s	439 kB	00:00
(12/42): guile22-2.2.7-2.amzn2023.0.3.x86_64.rpm	17 MB/s	6.4 MB	00:00
(13/42): infinipath-psm-3.3-26_g604758e_open.6.amzn2023.3.0.3.x86_64.rpm	3.2 MB/s	198 kB	00:00
(14/42): hwloc-libs-2.4.1-3.amzn2023.0.6.x86_64.rpm	24 MB/s	2.1 MB	00:00
(15/42): libfabric-1.14.0-2.amzn2023.0.2.x86_64.rpm	23 MB/s	860 kB	00:00
(16/42): kernel6.18-headers-6.18.51-120.162.amzn2023.x86_64.rpm	28 MB/s	1.6 MB	00:00
(17/42): libgfortran-14.2.1-7.amzn2023.0.2.x86_64.rpm	19 MB/s	963 kB	00:00
(18/42): libibumad-48.0-1.amzn2023.0.1.x86_64.rpm	896 kB/s	29 kB	00:00
(19/42): libmpc-1.2.1-2.amzn2023.0.2.x86_64.rpm	2.3 MB/s	62 kB	00:00
(20/42): libpsm2-11.2.86-8.amzn2023.0.2.x86_64.rpm	8.6 MB/s	206 kB	00:00
(21/42): libquadmath-14.2.1-7.amzn2023.0.2.x86_64.rpm	6.8 MB/s	210 kB	00:00
(22/42): libquadmath-devel-11.5.0-5.amzn2023.0.5.x86_64.rpm	5.5 MB/s	176 kB	00:00
(23/42): librdmacm-48.0-1.amzn2023.0.1.x86_64.rpm	2.3 MB/s	73 kB	00:00
(24/42): libtool-ltdl-2.4.7-1.amzn2023.0.3.x86_64.rpm	1.3 MB/s	38 kB	00:00

```
[ec2-user@ip-172-31-20-184 lab8]$ mpirun --version
mpicxx -o HelloMPI HelloMPI.c
mpirun --oversubscribe -np 4 ./HelloMPI
mpirun (Open MPI) 4.1.2
```

```
Report bugs to http://www.open-mpi.org/community/help/
Just a normal process From rank 3 machine ip-172-31-20-184.ec2.internal
Just a normal process From rank 1 machine ip-172-31-20-184.ec2.internal
Just a normal process From rank 2 machine ip-172-31-20-184.ec2.internal
Hello World! From rank 0 machine ip-172-31-20-184.ec2.internal
[ec2-user@ip-172-31-20-184 lab8]$
```

```
[ec2-user@ip-172-31-20-184 lab8]$ mpicxx -o messages1 messages1.cc
mpirun --oversubscribe -np 4 ./messages1
mpicxx -o message2 message2.cc
mpirun --oversubscribe -np 4 ./message2
Just a normal process From rank 0 machine ip-172-31-20-184.ec2.internal
Just a normal process From rank 2 machine ip-172-31-20-184.ec2.internal
SEnding message to computer 3 from computer 1
in computer 3 the value of y is 30
Just a normal process From rank 2 machine ip-172-31-20-184.ec2.internal
Just a normal process From rank 0 machine ip-172-31-20-184.ec2.internal
SEnding message to computer 3 from computer 1
in computer 3 the value of y is printed
0 10 20 30 40 50 60 70 80 90 [ec2-user@ip-172-31-20-184 lab8]$ |
```

Open MPI 4.1.2 was installed on a new EC2 instance (Amazon Linux 2023, 2 vCPUs) together with gcc and gcc-c++. The three in-class programs from the SE4060-MPI-Part1 repository were compiled with mpicxx and run with 4 processes. HelloMPI prints one line from every rank. In messages1.cc rank 1 sends $x = 30$ to rank 3 with MPI_Ssend and rank 3 receives it into y with MPI_Recv; ranks 0 and 2 only print their rank. message2.cc is the same, but an array of 10 integers (0, 10, ... 90) is sent in a single message.

The order of the lines changes from run to run because the processes run at the same time.